

Internship Prerequisite Mapping & Phased Pre-Study Plan

Computer Engineering | Smart Device Prototype Manufacturing (Maker) Internship —
Design · Output · Control (v8 phased competency design)

Pipeline: Design (Fusion 360) → Output (3D Printer) → Control (Arduino + C)

Table 1 — Internship Subject Matching Table (Smart Device Prototype Manufacturing Standard)

Area	Class Hours	On-site Work Content	Matching Subject (tentative)	Grade	Content Covered by Subject	Status
Embedded Control (Firmware)	10h	Sensor/actuator control logic, firmware writing	C프로그래밍 (비교과 = non-credit)	1st yr	Variables, control statements, functions, arrays, pointers, 임베디드 C basics	✔ Done — no further prep needed
Embedded Control (HW)	10h	Arduino circuit setup, sensor integration, serial comms	마이크로프로세서 · 임베디드시스템 (비교과)	2nd yr	MCU structure, GPIO, ADC/PWM, interrupts, communication	📦 Prep pack exists (arduino-internship-prep/) — in progress
3D Design (CAD)	10h	Part/housing modeling, assembly & tolerance design	컴퓨터그래픽스 / CAD 기초 (비교과)	2–3rd yr	3D shape modeling, parametric design	🆕 New plan below
Digital Manufacturing (Output)	10h	3D printer slicing, output, post-processing	— no matching regular course; self-study required —	–	–	🆕 New plan below
AI Utilization	10h	Document/code/report-writing support	프롬프트엔지니어링활용 (교과 = credit course)	1–2nd yr	Generative AI usage methods, prompt-design fundamentals	🆕 New plan below

Total pre-study load: 5 areas × 10h = 50 hours, split into 5×2h sessions per area (25 sessions total).

What this package contains

Since 임베디드 제어(펌웨어) / C프로그래밍 is already completed, four prep tracks remain. Each one follows the same format as your existing Arduino pack: 5 sessions × 2 hours, a `notes.md` + hands-on practice per session, and a `progress-tracker.md` checklist.

Folder	Area	Matches table row
arduino-internship-prep/ (already have this)	Embedded Control (HW)	Row 2
cad-design-prep/	3D Design (CAD)	Row 3

3d-printing-prep/	Digital Manufacturing (Output)	Row 4
ai-prompt-engineering-prep/	AI Utilization	Row 5

Work through them in **pipeline order** — this matches how the actual internship workflow runs:

1. cad-design-prep/ (Design in Fusion 360)
↓
2. 3d-printing-prep/ (Output the design)
↓
3. arduino-internship-prep/ (Control – you're already partway through this)
↓
4. ai-prompt-engineering-prep/ (cross-cutting – use anytime, doesn't block the others)

C프로그래밍 isn't listed above since it's already ✔ done — it was the prerequisite that makes the Arduino pack's C syntax make sense.